Jordan Canonical Form

Problem sheet for the Day 11 class · CSIR NET / GATE previous year questions and exam-pattern problems

Name: Date: Score: / 10

Instructions. 10 problems from CSIR NET, GATE (MA) and IIT JAM. A tag with an exam and year marks an actual previous-year question; a tag reading *Exam pattern* marks a problem written in the same style. MCQ: one correct option. MSQ: one or more correct options. NAT: write the number. Suggested time: 30 minutes. **The answer key is posted 24 hours after this sheet.**

1. GATE 2008 MCQ Let $T : \mathbb{R}^4 \to \mathbb{R}^4$ be the linear map satisfying $T(e_1) = e_2,\ T(e_2) = e_3,\ T(e_3) = 0,\ T(e_4) = e_2$, where $\{e_1, e_2, e_3, e_4\}$ is the standard basis of $\mathbb{R}^4$. Then

(A) T is idempotent
(B) T is invertible
(C) $\mathrm{Rank}\, T = 3$
(D) T is nilpotent

2. GATE 2013 MCQ Let V be a vector space of dimension $m \geq 2$. Let $T : V \to V$ be a linear transformation such that $T^{n+1} = 0$ and $T^n \neq 0$ for some $n \geq 1$. Then which of the following is necessarily TRUE?

(A) $\mathrm{Rank}(T^n) \leq \mathrm{Nullity}(T^n)$
(B) $\mathrm{trace}(T) \neq 0$
(C) T is diagonalizable
(D) $n = m$

3. Exam pattern MCQ Up to similarity, the number of nilpotent 5×5 complex matrices is

(A) 5
(B) 6
(C) 7
(D) 8

4. Exam pattern MCQ Let A be a 6×6 complex matrix with characteristic polynomial $(x-2)^4(x-1)^2$, minimal polynomial $(x-2)^2(x-1)$, and $\dim \ker(A - 2I) = 3$. Then $\mathrm{rank}(A - 2I)$ is

(A) 2
(B) 3
(C) 4
(D) 5

5. Exam pattern NAT The number of similarity classes of 4×4 complex matrices with minimal polynomial $(x-1)^2(x-2)$ is _____.

6. CSIR NET Dec 2015 MSQ Consider the matrices $A = \begin{pmatrix} 2 & 2 & 1 \\ 0 & 2 & -1 \\ 0 & 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

Then

(A) A and B are similar over the field of rational numbers $\mathbb{Q}$

(B) A is diagonalizable over the field of rational numbers $\mathbb{Q}$

(C) B is the Jordan canonical form of A

(D) The minimal polynomial and the characteristic polynomial of A are the same

7. CSIR NET June 2016 MSQ Let T be a $n \times n$ matrix with the property $T^n = 0$. Which of the following is/are true?

(A) T has n distinct eigenvalues

(B) T has one eigenvalue of multiplicity n

(C) 0 is an eigenvalue of T

(D) T is similar to a diagonal matrix

8. CSIR NET June 2017 MSQ Let V be the vector space of polynomials of degree at most 3 in a variable x with coefficients in $\mathbb{R}$. Let $T = d/dx$ be the linear transformation of V to itself given by differentiation. Which of the following are correct?

(A) T is invertible

(B) 0 is an eigenvalue of T

(C) There is a basis with respect to which the matrix of T is nilpotent

(D) The matrix of T with respect to the basis $\{1, 1+x, 1+x+x^2, 1+x+x^2+x^3\}$ is diagonal

9. Exam pattern MSQ Let N be a 4×4 complex nilpotent matrix with $\operatorname{rank} N = 2$. Which of the following are true?

(A) $N^2 = 0$ is possible

(B) $N^3 = 0$

(C) $\operatorname{rank} N^2 = 1$ is possible

(D) $\operatorname{nullity} N^2 = 2$

10. Exam pattern MSQ Let $A \in M_3(\mathbb{C})$ satisfy $A^3 = A^2$ and $A \neq A^2$. Which of the following are true?

(A) A is not diagonalizable

(B) A is nilpotent

(C) A^2 is diagonalizable

(D) $\operatorname{rank} A \geq 1$